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have given the rest to Denny. The number of lollipops he has given to Denny must have been $20 - 12 = 8$ lollipops. The answer is 8.

Q: Shawn has five toys. For Christmas, he got two toys each from his mom and dad. How many toys does he have now?

A: He has 5 toys. He got 2 from his Mom, so after that, he has $5 + 2 = 7$ toys. Then he got 2 more from Dad, so in total, he has $7 + 2 = 9$ toys. The answer is 9.

Q: There were nine computers in the server room. Five more computers were installed each day, from

Monday to Thursday. How many computers are now in the server room?

A: There are 4 days from Monday to Thursday. 5 computers were added each day. That means in total $4 * 5 =$

20 computers were added. There were 9 computers in the beginning, so now there are $9 + 20 = 29$ computers.

The answer is 29.

Q: Michael had 58 golf balls. On Tuesday, he lost 23 golf balls. On Wednesday, he lost 2 more. How many

What golf balls did he have at the end of Wednesday?

A: Michael initially had 58 balls. He lost 23 on Tuesday, so after that, he has $58 - 23 = 35$ balls. On

Wednesday he lost 2 more so now he has $35 - 2 = 33$ balls. The answer is 33.

Q: Olivia has \$23. She bought five bagels for \$3 each. How much money does she have left?

A: She bought 5 bagels for \$3 each. This means she spent 5

Q: When I was 6 my sister was half my age. Now I'm 70 how old is my sister?

A:

Output:

67

4. Generated Knowledge Prompting:

The produced information may help you arrive at the correct solution for a certain problem. It employs a pre-trained language model and a prompt template as the initial step to produce knowledge and is described in the work by Liu et al. published in 2022. To guide the model in the proper path, we give precise instructions and examples.

This method is applicable to many different jobs and prompts, and it may be improved by adding additional sources and APIs to expand the model's knowledge. We may receive better response suggestions and assess them to get the most probable right answer by creating information and enhancing the question with it. An interesting field of study that has the potential to

completely transform a wide range of applications and businesses is the notion of employing language models to produce knowledge for certain tasks.

Example:

Input: Greece is larger than Mexico.

Knowledge: Greece is approximately 131,957 sq km, while Mexico is approximately 1,964,375 sq km, making Mexico 1,389% larger than Greece.

Input: Glasses always fog up.

Knowledge: Condensation occurs on eyeglass lenses when water vapor from your sweat, breath, and ambient humidity lands on a cold surface, cools, and then changes into tiny drops of liquid, forming a film that you see as fog. Your lenses will be relatively cool compared to your breath, especially when the outside air is cold.

Input: A fish is capable of thinking.

Knowledge: Fish are more intelligent than they appear. In many areas, such as memory, their cognitive powers match or exceed those of 'higher' vertebrates including non-human primates. Fish's long-term memories help them keep track of complex social relationships.

Input: A common effect of smoking lots of cigarettes in one's lifetime is a higher than normal chance of getting lung cancer.

Knowledge: Those who consistently averaged less than one cigarette per day over their lifetime had nine times the risk of dying from lung cancer than never smokers. Among people who smoked between one and 10 cigarettes per day, the risk of dying from lung cancer was nearly 12 times higher than that of never smokers.

Input: A rock is the same size as a pebble.

Knowledge: A pebble is a clast of rock with a particle size of 4 to 64 millimetres based on the Udden-Wentworth scale of sedimentology. Pebbles are generally considered larger than granules (2 to 4 millimetres diameter) and smaller than cobbles (64 to 256 millimetres diameter).

Input: Part of golf is trying to get a higher point total than others.

Output:

The objective of golf is to play a set of holes in the least number of strokes. A round of golf typically consists of 18 holes. Each hole is played once in the round on a standard golf course. Each stroke is counted as one point, and the total number of strokes is used to determine the winner of the game.

5. Program-aided Language Model (PAL)

The intermediate phases of program-aided language models